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Unconventional hydrocarbon recovery

This chapter describes how to model hydrocarbon recovery from unconventional sources such as coal bed methane and shale gas.

Coal bed methane model

x	Eclipse 100
x	Eclipse 300

The naturally occurring methane within coal seams can in some cases be economically produced using oil field technology. In these “Coal Bed Methane” projects, wells are drilled into the coal seam to produce gas. Considerable reserves of this “unconventional” natural gas are present in many parts of the world.

The coal beds are naturally fractured systems with the gas adsorbed into the coal matrix. Primary production occurs by initially de-watering the natural fractures and hence reducing the pressure in the fracture system. The reduced pressure in the fractures allows gas desorption from the surface of the coal to the fracture. Gas diffuses from the bulk of the coal towards the fracture surface.

For dual porosity the coal bed methane model uses a modified Warren and Root model to describe the physical processes involved in a typical coal bed methane project.

In Eclipse 100 it is possible to introduce a second gas, known as a solvent, for enhanced Coal Bed Methane projects, while in Eclipse 300 a full compositional treatment is possible.

CAUTION: You should be aware that certain entities, corporations, individuals or organizations may or do hold applicable patents with regard to enhanced recovery of coal bed gases through injection of both inert and non-inert gases. You are strongly encouraged to investigate if any proposed field applications are subject to these patents.

Single porosity model

In Eclipse 300 it is possible to activate an instant sorption model that can be used in single porosity mode. With this model the amount of adsorbed gas is immediately in equilibrium with the free gas phase.

Dual porosity model

The dual porosity model consists of two interconnected systems representing the coal matrix and the permeable rock fractures. To model such systems, two simulation cells are associated with each block in the geometric grid, representing the coal matrix and fracture volumes of the cell. In Eclipse, the properties of these blocks may be independently defined. In an Eclipse dual porosity run, the number of layers in the Z-direction should be doubled. Eclipse associates the first half of the grid (the first `NDIVIZ` /2 layers) with the matrix blocks, and the second half with the fractures. In such runs `NDIVIZ` must therefore be even; Eclipse checks that this is the case.

Unlike a dual porosity oil reservoir model, in which the matrix has both an associated pressure and an oil saturation, only the gas concentration in the coal is tracked. In the fracture system, however, the standard flow equations are solved.

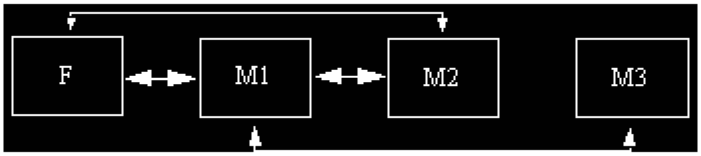
The `DUALPERM` option should only be used if the instant adsorption model in Eclipse 300 is activated (`CBMOPTS`) or if the time-dependent adsorption model is used with multi-porosity. With the time-dependent sorption model, one of the submatrix porosities will then represent the pore volume of the matrix having conductivity to neighbor matrix grid blocks, while grid cells assigned to be coal will get a zero transmissibility.

Note: In a coal bed methane model the pore volume of the matrix cells has a different interpretation than for an ordinary dual porosity run; it gives the coal volume of the cell, using the time-dependent sorption model. By default the porosity is set to unity minus the porosity of the fracture. The cell bulk volume multiplied by the porosity then equals the coal volume. It is possible to use the keyword `ROCKFRAC` instead of the porosity to set the coal volume in Eclipse 300.

Note: By default the input permeabilities in the fracture are scaled by the porosities. To turn this off use the keyword NODPPM.

Triple porosity model

In Eclipse 300, the triple porosity model, TRPLPORO, can be activated together with the Coal Bed Methane option. The triple porosity model offers an extra connection between the second submatrix (M2) and the fracture (F) and, if using four porosities, from the “isolated” third submatrix (M3) to the first submatrix (M1) as shown in the diagram below.



This enables the modeling of conventional flow in micro pores or mini fractures within the coal matrix together with diffusion of coal gas as formulated by the time-dependent sorption model. Different setups are possible depending on the setting of adsorbed gas in the submatrix. For the transmissibility and/or diffusivity between the M2-F connection, SIGMAV values are allowed to be assigned to the fracture grid cells. Note that to model dual permeability, the connection transmissibilities could be set by PERMMF instead of using the matrix PERMX values, but K-layer indices in the PERMMF case only run over matrix porosities while for the triple porosity model the SIGMAV values include the fracture porosity. For more information see "Triple Porosity" and also consider the example data set CBMTRPL.DATA.

Multi porosity model

In Eclipse 300 a multi porosity model can be activated, enabling a detailed study of the transient behavior in the matrix. See "Multi porosity".

Adsorption/diffusion models in Eclipse 100

The adsorbed concentration on the surface of the coal is assumed to be a function of pressure only, described by a Langmuir isotherm. The Langmuir isotherm is input as a table of pressure versus adsorbed concentrations. Different isotherms can be used in different regions of the field. See keyword COALNUM.

The diffusive flow of gas from the coal matrix is given by:

$$F_g = \mathrm{DIFFMF} \cdot D_c \cdot \left(\mathrm{GC}_b - \mathrm{GC}_s\right) \quad \text{Eq. 7.1}$$

where

F_g = gas flow

DIFFMF = matrix-fracture diffusivity

D_c = diffusion coefficient

GC_b = bulk gas concentration

GC_s = surface gas concentration.

GC_s is a function of fracture pressure given by the LANGMUIR table. The diffusion coefficient D_c is supplied using the DIFFCOAL keyword.

The diffusivity is given by:

$$\text{DIFFMF} = \text{DIFFMMF} \cdot \text{VOL} \cdot \sigma \tag{Eq. 7.2}$$

where

DIFFMMF is the multiplying factor input using DIFFMMF (default = 1.0)

VOL is the cell coal volume

σ is the factor to account for the matrix-fracture interface area per unit volume.

Sigma is a shape factor which accounts for the matrix fracture interface area per unit volume of coal matrix. For pseudo-steady-state flow in the coal matrix, the gas desorption time (time) is

$$\tau = \frac{1}{\sigma \cdot D_c \cdot DIFFMMF}$$

If the surface gas concentration is greater than the bulk gas concentration, gas may be re-adsorbed into the coal. In this case the flow of gas is treated slightly differently, as shown below. It is also possible to control the rate of re-adsorption and, in the limiting case, to prevent re-adsorption.

$$F_g = \text{DIFFMF} \cdot D_c \cdot S_g \cdot \text{RF} \cdot \left(\text{GC}_b - \text{GC}_s\right) \tag{Eq. 7.3}$$

where

S_g is the gas saturation in the fracture

RF is the re-adsorption factor, input using the DIFFCOAL keyword (default = 1.0).

Gas in the fracture will not contact the entire surface area of the coal if water is also present in the fracture. Hence the flow of gas into the coal is reduced by S_g to represent the reduced contact area.

The RF factor can be used to control the rate of re-adsorption. Setting RF = 0 prevents gas re-adsorption.

Adsorption models in Eclipse 300

The sorption models that can be used are:

- Table input, the adsorption capacities for each component are input by the LANGMUIR keyword.
- The Extended Langmuir isotherm. See the keyword LANGMEXT.

Table input

Using the keyword LANGMUIR, the adsorption capacities of each component are given versus pressure. Table lookup is done using the partial pressure, in order to determine the adsorption capacity of each component. The first row of the table is recommended to be at zero (partial) pressure, defining the minimum values to be zero. Thus if the component is not present the adsorption capacity is set to zero.

The partial pressures, used to look up the adsorption capacities, are scaled with corresponding LANGMPL values if this keyword is present.

Extended Langmuir isotherm

The Extended Langmuir isotherm *(Arri et al, 1992)* can be used to describe the coal sorption for the different components. See keyword LANGMEXT.

The adsorption capacity is a function of the pressure and the free gas phase composition. For each component, two parameters need to be input: the Langmuir volume constant V_i and the Langmuir pressure constant P_i. These parameters are typically determined from experiments. Different isotherms can be used in different regions of the field. See keyword COALNUM.

The multi-component adsorption capacity is calculated by:

$$L(p, \alpha, y_1, y_2, \ldots)_i = \theta \frac{P_s}{RT_s} \left(V_i \frac{y_i \frac{\alpha p}{P_i}}{1 + \sum_{\mathrm{j=1}}^{\mathrm{N_{comps}}} y_j \frac{\alpha p}{P_j}} \right)$$

where

θ = Scaling factor

P_s = Pressure at standard conditions

R = Universal gas constant

T_s = Temperature at standard conditions

V_i = Langmuir volume constant for component i

P_i = Langmuir pressure constant for component i

y_i = Hydro carbon mole fraction in gas phase for component i

p = Pressure

α = Pressure multiplier as given by LANGMPL

For the special case of a single component, the Extended Langmuir isotherm is identical to the usual Langmuir isotherm giving the storage capacity as a function of pressure only:

$$L(p, \alpha) = \theta \frac{P_s}{RT_s} \left(V \frac{\frac{\alpha p}{P}}{1 + \frac{\alpha p}{P}} \right) \quad \text{Eq. 7.4}$$

where **V** is the maximum storage capacity for the gas, referred to as the Langmuir volume constant, and **P** is the Langmuir pressure constant. The constants used in the Extended Langmuir formulation can hence be estimated from a series of single-component gas experiments.

Since the Langmuir volume constant V is given as surface volume over coal weight, the simulator translates this to moles over coal weight using the ideal gas law (with the term $\frac{P_s}{RT_s}$).

It is possible to scale the adsorption capacity by a factor θ for each cell in the grid with the LANGMULT keyword. Typically this can be used to account for differences in ash or moisture contents. To apply a different scaling for each component the keyword LANGMULC can be used.

Note that LANGMULT and LANGMULC cannot be used together.

Time dependent diffusion in Eclipse 300

The diffusive flow between the matrix and the fracture is given by

$$F_i = DIFFMF \cdot D_{c,i} \cdot S_g \cdot RF_i \cdot \left(m_i - \rho_c L_i\right) \qquad \text{Eq. 7.5}$$

where

m$_i$	= Molar density in the matrix coal.
$DIFFMF$	= Matrix fracture (or multi porosity) diffusivity.
ρ_c	= Rock density (Coal density), specified using ROCKDEN.
$D_{c,i}$	= Diffusion coefficient (coal) component i, specified using DIFFCBM.
RF_i	= Readsorption factor component i, specified using RESORB.
S_g	= Gas saturation. For desorption, a value of unity is used.
$\rho_c L_i$	= Equilibrium molar density of adsorbed gas, specified using LANGMUIR or specified using ROCKDEN for the coal density ρ_c with LANGMEXT for L_i

As can be seen by the flow equation and as noted in the above table the density of the coal needs to be specified, as this is multiplied by the Langmuir adsorption capacity. The term $\rho_c L_i$ gives the equilibrium adsorbed molar density. The coal density ρ_c is entered with the keyword ROCKDEN. When using the LANGMUIR keyword to specify the sorption, the coal density is not used as the input is then given as a concentration (that is $\rho_c L_i$ is entered directly).

For adsorption, the area of the gas in contact with the coal is accounted for by scaling with the gas saturation. (It is possible to use a value of unity instead with item 93 of the OPTIONS3 keyword)

The matrix-fracture diffusivity is given by

$$\mathrm{DIFFMF} = \mathrm{DIFFMMF} \cdot \mathrm{VOL} \cdot \sigma \qquad \text{Eq. 7.6}$$

where

DIFFMMF	is the multiplying factor input using DIFFMMF (default = 1.0)
VOL	is the coal volume
σ	is the factor to account for the matrix-fracture interface area per unit volume.

Often the component's sorption time is a quantity that is easier to obtain than the diffusion coefficients. For desorption the flow can be written as

$$F_i = \frac{VOL}{\tau_i} \cdot \left(m_i - \rho_c L_i\right) \qquad \text{Eq. 7.7}$$

where

$$\tau_i = \frac{1}{D_{c,i} \cdot DIFFMMF \cdot \sigma}$$

is called the sorption time. The parameter controls the time lag before the released gas enters the coal fracture system. The sorption times are given by the diffusion coefficients, DIFFCBM, and the matrix-fracture interface area, SIGMA, together with the multiplying factor DIFFMMF. If the sorption times are known a value of unity can be assigned to σ and DIFFMMF. The diffusion coefficients can then be assigned to the reciprocal of the sorption times.

The time dependent diffusive flow can be used with

- Dual porosity
- Multi porosity

It is possible to have connections between coal cells having the same coal region numbers. For connections between cells having different adsorption properties it is assumed that a pore volume cell is placed in between.

Diffusion between matrix-matrix cells

Using the multi-porosity option, a coal cell can connect to a matrix pore volume cell. The diffusion between two cells I and J is computed as:

$$F_i^{I \to J} = DIFF^{IJ} \cdot D_{c,i} \cdot S_g \cdot RF_i \cdot \left(m_i^I - \rho_c L_i^J\right)$$

with the assumption that I is the coal cell. Note that the innermost sub-matrix grid cell is used when calculating the diffusivity.

$$\mathrm{DIFF}^{IJ} = \mathrm{DIFFMMF}^{I} \cdot \mathrm{VOL}^{I} \cdot \sigma^{I}$$

where the index I corresponds to the index of the sub-matrix grid-cell I. If cell I corresponds to a pore volume cell, the volume used is the pore volume. The diffusivity multipliers and sigma factors are specified by the keywords DIFFMMF and SIGMAV, while the volumes are input by the appropriate ROCKFRAC or PORO keywords.

For pre-2012.1 versions of Eclipse, the volume used also corresponds to the coal volume for inner matrix sub-grid cells. To obtain this behavior, use item 278 in the OPTIONS3 keyword. For some combinations of the coal options, as for example with the Triple Porosity option, this might still be useful. The sigma factor then needs to be altered to account for the two connections from the coal volume.

Diffusion between coal cells

Using the multi-porosity option, a coal cell can connect to another coal cell. This is allowed if the grid cells have the same coal region number, thus having the same sorption properties. The potential for diffusion between two coal cells I and J are computed by using the molar density, m_i^I and m_i^J, in the coal volumes

$$F_i^{I \to J} = DIFF^{IJ} \cdot D_{c,i} \cdot \left(m_i^I - m_i^J\right)$$

The innermost sub-matrix grid cell is used when calculating the diffusivity.

$$\mathrm{DIFF}^{IJ} = \mathrm{DIFFMMF}^{I} \cdot \mathrm{VOL}^{I} \cdot \sigma^{I}$$

Instant sorption in Eclipse 300

If the sorption time approaches zero, the adsorbed gas will be in instant equilibrium with the free gas phase. The keyword CBMOPTS is used to activate the instant sorption model.

The instant sorption model can be used with

- Single porosity
- Dual porosity
- Multi-porosity

With this model every cell is assumed to contain a pore volume, the remaining fraction of the bulk volume being taken as the coal volume. For dual and multi-porosity models this is usually not appropriate as each pore system will add up the rock volume. In this case an average rock volume over the pore systems is calculated and partitioned among the connected pore systems. The default behavior can be changed by using the keyword ROCKFRAC to specify the fraction of the bulk volume of each simulation grid cell making up the coal volume. The sum of the porosities and the rock fractions over the connected pore systems should usually be unity.

Note, when using the instant sorption model, it is not possible to assign a tracer to the adsorbed components.

Rock compaction

There are several models available in order to model the compression and expansion of the pore volumes and the effect on permeability for a coal bed methane reservoir.

A Palmer-Mansoori model (*Palmer and Mansoori, 1998*) is available using the ROCKPAMA keyword.

Options for the ROCKPAMA and ROCKPAME models can be activated by using CBMOPTS items 4-9.

For ROCKPAMA, this includes modeling the sorption induced strain term as a function of the actual adsorbed gas concentration, while by default this is estimated using an instant adsorption assumption. This option in CBMOPTS is important if the diffusion process of the sorption process is so that the standard Palmer-Mansoori model would predict swelling/shrinkage due to the instant adsorption assumption, without the support of gas exchange. For cases where the standard Palmer-Mansoori model predicts a reduction in pore volume for increased pressure, this needs to be supported by gas adsorbing into the coal.

It is possible to tabulate the sorption induced strain term versus the coal gas concentration using the keywords TPAMEPS and TPAMEPSS. This also means that ROCKPAMA can be used for multi-components.

For the rock models ROCKPAMA and ROCKPAME it is possible to output the number of passes through the rebound pressure for each grid cell. The rebound pressure is when the pressure derivative of the pore volume changes sign. This output is made possible using NPMREB with RPTSCHED and RPTRST, and activating the output using CBMOPTS item 4.

Eclipse 300

Another extension for multi-component gases is available by specifying the ROCKPAME keyword. This enables the modeling of coal swelling and shrinkage as a function of both composition and pressure, but assumes an instant adsorption process for the sorption induced strain term.

For more information see "Rock compaction".

The effect of compaction is only modeled in the pore volume. The coal volume of a simulation cell and its adsorbed contents are related to the surface gas volume per unit coal volume. For example, if ROCKTAB or

ROCKTABH keyword is defined, the pore volume multiplier (as a function of block pressure – same for fracture and matrix cell) is not applied to the matrix volume of coal cells.

Modeling compositional effects with Eclipse 100

Improved recovery of methane from coal beds can be achieved by a secondary recovery scheme involving the injection of a second gas, typically nitrogen or carbon dioxide. There are two main mechanisms that aid recovery of the coal’s methane content. Firstly, by reducing the partial pressure of the methane present in the fracture, the diffusion from the coal is enhanced. The second mechanism is the competitive adsorption of carbon dioxide into the coal matrix. Molecules of CO_2 preferentially sorb on to the surface adsorption sites, displacing methane.

Eclipse allows the modeling of a range of secondary recovery processes by introducing a second gas, known as a solvent, which can be employed to model either nitrogen or CO_2 injection. Two models can be used to describe the adsorption process:

Adsorption model 1: COALADS keyword

The solvent/gas adsorption model is based on three input tables:

- Langmuir isotherm for pure gas, typically methane.
  The table defines the gas surface concentration versus pressure.
- Langmuir isotherm for pure solvent, CO_2 say.
  The table defines the solvent surface concentration versus pressure.
- Table of the relative adsorption of gas and solvent as a function of the composition of the fracture gas.

Then:

Surface concentration of gas = $C_g(P) \cdot F_g\left(f_g\right)$

Surface concentration of solvent = $C_s(P) \cdot F_s\left(f_g\right)$

Where

$C_g(P)$ is the adsorption of pure gas at pressure P (LANGMUIR keyword)

$C_s(P)$ is the adsorption of pure solvent at pressure P (LANGSOLV keyword)

F_g is the gas relative adsorption (COALADS keyword)

F_s is the solvent relative adsorption

f_g is the fraction of gas in the fracture $= S_g/\left(S_g + S_{\text{solv}}\right)$

Adsorption model 2: COALPP keyword

The second adsorption model is based on the partial pressures of the component gases. Again, three input tables are required:

- Langmuir isotherm for pure gas, typically methane.
  The table defines the gas surface concentration versus pressure.

- Langmuir isotherm for pure solvent, CO_2 for example.
  The table defines the solvent surface concentration versus pressure
- Table of the relative adsorption of gas and solvent as a function of the composition of the fracture gas.

Then:

Surface concentration of gas = $C_g(P') = C_g(P \cdot F_g(f_g))$

Surface concentration of solvent = $C_s(P') = C_s(P \cdot F_s(f_g))$

Where

$C_g(P')$ is the adsorption of gas at partial pressure P' (LANGMUIR keyword)

$C_s(P')$ is the adsorption of solvent at partial pressure P' (LANGSOLV keyword)

F_g is the gas relative adsorption (COALPP keyword)

F_s is the solvent relative adsorption

fg is the fraction of gas in the fracture = $S_g/(S_g + S_{solv})$

Note: Either the COALADS or COALPP keyword must be specified when the secondary recovery scheme, that is coal and solvent, is used.

Using the Eclipse 100 Coal Bed Methane Model

Single gas mode

The model is activated by specifying the keyword COAL in the RUNSPEC section. The dual porosity model should also be activated with the RUNSPEC keyword DUALPORO, and the program should be run in gas-only or gas-water mode (OIL may not be specified in RUNSPEC).

Grid data

The first NDIVIZ /2 layers of the model represent the coal matrix, and the second NDIVIZ /2 layers represent the fracture system. It is possible to have active fracture cells with no corresponding matrix cell; however, all active matrix cells require a corresponding active fracture cell. The data required for the coal matrix consists of:

Geometry

Either DX / DY / DZ or ZCORN / COORD

Porosity

This is used to calculate the coal volume, and is taken to represent the fraction of the volume occupied by coal. (The porosity can be defaulted, in which case it is set to 1 - fracture porosity.)

Net-to-Gross

Optional, defaults to 1.0

Coal region number

COALNUM. This is optional and defaults to 1 for all the matrix cells. Note that for a matrix cell given a zero value, the cell is taken as a normal pore volume cell and porosities and permeabilities need to be assigned. The number of coal regions should be set by item 6 of REGDIMS (default =1).

The grid block coal volumes can be output using the 'PORV' mnemonic in the RPTGRID keyword. Note that region and field coal volumes are printed out on the fluid-in-place balance sheets (mnemonic 'FIP' in RPTSOL and RPTSCHED).

The matrix-fracture connection factor, sigma, needs to be supplied using either the SIGMA or SIGMAV keywords.

Properties data

Gas and water PVT and relative permeability data need to be supplied in the usual way.

In addition the following keywords must be supplied:

Keyword	Description
LANGMUIR	Inputs tables of surface gas concentration versus pressure.
DIFFCOAL	Inputs the diffusion coefficient and the re-adsorption factor.

It is possible to scale the tables of surface gas concentration versus pressure on a cell-by-cell basis using the MLANG keyword. This keyword is optional, and if it is omitted the raw table data is used.

Equilibration

If the model is equilibrated using the EQUIL keyword, no extra data is required. In this case the initial coal gas concentration is set to the equilibrium concentration at the prevailing fracture pressure. Two optional keywords are available to set the initial coal gas concentration manually:

Keyword	Description
GASCONC	Inputs the coal gas concentration on a block-by-block basis.
GCVD	Inputs the coal gas concentration as a function of depth.

Care should be taken when using non-equilibrium starts. Eclipse prints a warning if the coal gas concentration exceeds the equilibrium value in any grid block.

For enumerated starts, one of the above two keywords must be present.

Summary data

Three extra SUMMARY section keywords are available to output the coal gas concentration on a field, region and block basis:

Keyword	Description
FCGC	Field coal gas concentration.
RCGC	Region coal gas concentration.
BCGC	Block coal gas concentration.
BPORVMOD	Block pore volume multiplier, rock compaction models.
BPERMMOD	Block permeability multiplier, rock compaction models.

Two gas mode - compositional effects

The second gas component is activated by entering the keyword SOLVENT in the RUNSPEC section.

Solvent properties and adsorption data

The properties of the solvent gas are supplied in an analogous manner to the gas phase, with keywords:

Keyword	Description
SDENSITY	Solvent surface density
PVDS	Solvent PVT tables
SSFN	Gas/solvent relative permeability data for flow in the fractures.

The adsorption process is described by two extra keywords:

Keyword	Description
LANGSOLV	Langmuir isotherm for pure solvent
COALADS or COALPP	Relative adsorption data for gas and solvent.

The LANGSOLV data can be scaled using the MLANGSLV keyword in a manner analogous to the MLANG keyword.

Either the COALADS keyword or the COALPP keyword can be used to specify the relative adsorption data. If the COALADS keyword is used, the relative adsorption is specified as a table of relative adsorption factors. The alternative COALPP keyword inputs a table of pressure factors for the partial pressure model.

Equilibration

By default the initial solvent fraction is taken to be zero. It is possible to specify an initial solvent fraction using one of two keywords:

Keyword	Description
SOLVFRAC	Inputs the initial solvent fraction in the fracture system. This keyword should be used when the initial conditions are specified by equilibration
SSOL	Inputs the initial solvent saturation for enumerated starts.

The initial concentrations of gas and solvent in the coal are then calculated from the LANGMUIR, LANGSOLV and COALADS or COALPP data. Two optional keywords are available to set the initial coal solvent concentration manually:

Keyword	Description
SOLVCONC	Inputs the coal solvent concentration on a block-by-block basis
SCVD	Inputs the coal solvent concentration as a function of depth.

Notes

- If initial mobile gas exists in the model, the initial conditions are not quite in equilibrium, as the solvent and gas components have different densities. The equilibration calculation is based on the gas (methane) properties only.

- An initial gas saturation in the reservoir can be modeled by setting the connate gas saturation greater than zero. This has the advantage of being in initial equilibrium, and allows Eclipse to calculate the initial coal concentration based on the SOLVFRAC data alone.
- The connate gas saturation is specified by starting the gas saturation table (SGFN) at S_{gco}, and ending the water saturation table (SWFN) at 1.0 - S_{gco}.

  It is also possible to initialize the model by enumeration and to include a solvent saturation. This is achieved using the SSOL keyword in the SOLUTION section.

Solvent injection

Solvent can be injected by specifying a gas injection well. A separate keyword WSOLVENT can then be used to specify the fraction of the gas flow that is solvent.

Summary data

Refer to Coal bed methane model in the *Eclipse Reference Manual.*

Summary of keywords (Eclipse 100)

RUNSPEC

Restriction	Keyword	Description
	CBMOPTS	Options related to the coal bed methane model
	COAL	Specifies that the coal bed methane model is to be used.
Required for dual porosity runs. **Do not use** DUALPERM	DUALPORO	Specifies a dual porosity run.
Must be used to activate the solvent model if you are using the two gas mode.	SOLVENT	Activates solvent model.

GRID

Keyword	Description
COALNUM	Assign coal regions.
DIFFMMF	Modify the Matrix-Fracture diffusivity.
ROCKFRAC	Specify rock (coal) fractions of the bulk volume on a cell by cell basis.
RPTGRID	Control output from the GRID section: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.
RPTGRIDL	Control output from the GRID section for local refined grids: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.
RPTINIT	Control output from the GRID and EDIT sections to INIT file: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.
SIGMA	Specify the dual porosity SIGMA factor.
SIGMAV	Specify the SIGMA factor on a cell by cell basis.

PROPS

Keyword	Description
COALADS	Specify the gas/solvent relative adsorption data.
COALPP	Specify the gas/solvent partial pressure data.
DIFFCOAL	Define the gas diffusion data.
LANGMPL	Tables of pressure scaling factors for the adsorption capacity for each grid cell.
LANGMUIR	Tables of coal surface gas concentration.
LANGSOLV	Tables of coal surface solvent concentration.
MLANG	Maximum surface gas concentration.
MLANGSLV	Maximum surface solvent concentration.
PVDS	PVT properties of solvent.
RPTPROPS	Controls output of the PROPS section: DIFFC or DIFFCOAL outputs the diffusion data ENDPTS outputs the MLANG and MLANGSLV scaling data LANGMUIR, LANGSOLV or COALADS outputs the Langmuir isotherms and relative adsorption data.
SDENSITY	Solvent density data.
SSFN	Gas/Solvent relative permeability data.
TPAMEPS	Sorption induced strain values for the ROCKPAMA model.
TPAMEPSS	Sorption induced solvent strain values for the ROCKPAMA model.

SOLUTION

Keyword	Description
GASCONC	Initial coal gas concentration.
GCVD	Initial coal gas concentration versus depth.
RPTSOL	Controls the output from the SOLUTION section: • FIPSOL outputs solvent fluid in place reports • GASCONC outputs matrix grid block gas and solvent concentrations • SSOL outputs current solvent saturation.
SCVD	Initial solvent concentration versus depth.
SOLVCONC	Initial coal solvent concentration.
SOLVFRAC	Initial solvent fraction in the gas phase
SSOL	Initial solvent saturation.

SUMMARY

Refer to Coal bed methane in the *Eclipse Reference Manual.*

Note: Methane production (FMPR etc.) is identical to the gas production (FGPR etc.) unless the solvent option is active. In that case the gas production rate (FGPR) is the total gas + solvent production rate.

SCHEDULE

Keyword	Description
RPTRST	The restart file will contain the arrays "COALGAS", "COALSOLV", "GASSATC", "MLANG" and " MLANGSLV" whose meanings are documented in the *Eclipse File Formats Reference Manual*
RPTSCHED	Control the output from the SCHEDULE section: • FIPSOL outputs solvent fluid in place reports • GASCONC outputs matrix grid block gas and solvent concentrations • SSOL outputs current solvent saturation.
WSOLVENT	Sets the solvent fraction in gas injection wells.

Examples Eclipse 100

Basic test Eclipse 100

RUNSPEC

Example of a basic Eclipse 100 test in the RUNSPEC section:

```
RUNSPEC    ======================================
TITLE
                  Coal Bed Methane Basic Test
DIMENS
 8    8    2  /
DUALPORO
WATER
GAS
FIELD
COAL
EQLDIMS
 1  100    2    1   20 /
TABDIMS
 1    1   20   20    3    5 /

 3    1    0    0    0    1 /
WELLDIMS
 2   13    1    2 /
NUPCOL
 4 /
START
 26 'JAN' 1983   /
NSTACK
 20 /
FMTOUT
FMTIN
```

GRID

Example of a basic Eclipse 100 test in the GRID section:

```
GRID    ============================================================
EQUALS
 'DX'       75    /    PROPERTIES COMMON TO MATRIX AND FRACTURES
 'DY'       75    /
 'DZ'       30    /
 'TOPS'   4000    /
 'PERMZ'     0    /
 'DIFFMMF'  1.0   /
 'PORO'   0.01         4*  2  2  /  FRACTURE PROPERTIES (LAYER TWO)
 'PERMX' 500000    /                MATRIX POROSITY DEFAULTS TO
 'PERMY' 500000    /                1 - PORO(FRACTURE)
-- INACTIVE MATRIX CELLS ARE OK
 'NTG'  0.0  3 3 3 3  1 1 /
 'NTG'  1.0  3 3 3 3  2 2 /
/
RPTGRID
 'DX'  'DY'  'DZ'  'PERMX'  'PERMY'  'PERMZ'  'MULTX'
 'MULTY'  'MULTZ'  'PORO'  'NTG'  'TOPS'  'PORV'
 'DEPTH'  'TRANX'  'TRANY'  'TRANZ'  'KOVERD'  /
--SIGMA FOR 10 BY 10 BY 30 FT BLOCKS
SIGMA
 0.08   /
```

PROPS

Example of a basic Eclipse 100 test in the PROPS section:

```
PROPS   ============================================================
--COAL GAS DIFFUSION COEFFICIENT
DIFFCOAL
 0.2  /

LANGMUIR
 0.0       0.0
 1000.0    0.05
 2000.0    0.065
 3000.0    0.083
 4000.0    0.10
 6000.0    0.11
/

SWFN
  0.0    0.0   0.0
  0.2    0.0   0.0
  1.0    1.0   0.0
/

SGFN
  0.0    0.0   0.0
  0.1    0.0   0.0
  1.0    1.0   0.0
/

PVTW
  .0000000   1.00000    3.03E-06    .50000    0.00E-01 /

--   PGAS   BGAS   VISGAS
PVDG
      400    5.9   0.013
      800    2.95  0.0135
     1200    1.96  0.014
     1600    1.47  0.0145
     2000    1.18  0.015
     2400    0.98  0.0155
     2800    0.84  0.016
     3200    0.74  0.0165
     3600    0.65  0.017
     4000    0.59  0.0175
     4400    0.54  0.018
     4800    0.49  0.0185
     5200    0.45  0.019
     5600    0.42  0.0195 /
```

```
ROCK
 4000.00        .30E-05 /

DENSITY
 52.0000  64.0000   .04400 /

RPTPROPS
 'SOF2'   'SWFN'   'SGFN'   'PVTO'
 'PVTW'   'PVTG'   'DENSITY'  'ROCK'  'DIFFC'
 'LANGMUIR'  /
```

REGIONS

Example of a basic Eclipse 100 test in the `REGIONS` section

```
REGIONS   ===========================================================
EQUALS
 'FIPNUM' 1  1 8 1 8 1 1 /
 'FIPNUM' 2  1 8 1 8 2 2 /
 'FIPNUM' 3  2 2 2 2 2 2 /
/
```

SOLUTION

Example of a basic Eclipse 100 test in the `SOLUTION` section:

```
SOLUTION  ==========================================================
EQUIL
4000  3959  4000  0   0   0   0   0   0  /
RPTSOL
 'PRES'  'SWAT'  'SGAS'  'FIP=2'  'EQUIL'  'GASCONC'  /
```

SUMMARY

Example of a basic Eclipse 100 test in the `SUMMARY` section:

```
SUMMARY    ==========================================================
FPR
FGPR
FWPR
FCGC
RCGC
 1 2 3 /
BCGC
 2 2 1 /
/
RUNSUM
SEPARATE
```

SCHEDULE

Example of a basic Eclipse 100 test in the `SCHEDULE` section:

```
SCHEDULE   =========================================================
RPTSCHED
 'PRES'  'SWAT'  'SGAS'  'RESTART=2'  'FIP=2'
 'WELLS=2'  'CPU=2'  'NEWTON=2'  /
WELSPECS
'P'  'G'   8  8  4000  'GAS'  /
/
COMPDAT
'P'  8  8  2  2 'OPEN'  0  .000000  .5000  .00000   .0000  0.000E-01/
/
WCONPROD
'P'  'OPEN'  'WRAT'     1*         100.0      100000.00000
     1*          1*      200.000  0.000000   1*  0.00000000/
```

```
/
TSTEP
 1.0 9.0 90.0 9*100 6*1000
/
END
```

CO_2 injection example Eclipse 100

RUNSPEC

Eclipse 100 CO_2 injection example in the `RUNSPEC` section:

```
RUNSPEC    ======================================
TITLE
                  CO2 Injection Example
DIMENS
 8    8    2  /
DUALPORO
WATER
GAS
SOLVENT
COAL
FIELD
EQLDIMS
 1  100    2    1   20 /
TABDIMS
 1    1   20   20    2    5 /

 2    1    0    0    0    1 /
WELLDIMS
 2   13    1    2 /
NUPCOL
 4 /
START
 26 'JAN' 1983  /
NSTACK
 20 /
```

GRID

Eclipse 100 CO_2 injection example in the `GRID` section:

```
GRID      ============================================================
EQUALS
 'DX'       75    /     PROPERTIES COMMON TO MATRIX AND FRACTURES
 'DY'       75    /
 'DIFFMMF'  1.0   /
 'DZ'       30    /
 'PERMZ'     0    /
 'PORO'   0.005      1  8  1  8  2  2  /  FRACTURE PROPERTIES
 'PERMX' 500000    /
 'PERMY' 500000    /
 'NTG'  0.0  3 3 3 3   1 1 /
 'NTG'  1.0  3 3 3 3   2 2 /
-- 10 per cent dip in the X - direction.
 'TOPS'   4000     1 1   1 8  1 2   /
 'TOPS'   4007.5   2 2   1 8  1 2   /
 'TOPS'   4015     3 3   1 8  1 2   /
 'TOPS'   4022.5   4 4   1 8  1 2   /
 'TOPS'   4030     5 5   1 8  1 2   /
 'TOPS'   4037.5   6 6   1 8  1 2   /
 'TOPS'   4045     7 7   1 8  1 2   /
 'TOPS'   4052.5   8 8   1 8  1 2   /
/
SIGMA
 0.08   /
```

PROPS

Eclipse 100 CO_2 injection example in the PROPS section:

```
PROPS   ============================================================
DIFFCOAL
 20.0 1.0 /

LANGMUIR
   0.0   0.0
  257.   0.092
  528.   0.14
 1000.   0.20
/

--
-- Coal quality improves in the Y direction
--
EQUALS
 'MLANG'     5   1 8  1 1  1 1 /
 'MLANG'     7   1 8  2 2  1 1 /
 'MLANG'     9   1 8  3 3  1 1 /
 'MLANG'    12   1 8  4 4  1 1 /
 'MLANG'    15   1 8  5 5  1 1 /
 'MLANG'    20   1 8  6 6  1 1 /
 'MLANG'    30   1 8  7 7  1 1 /
 'MLANG'    35   1 8  8 8  1 1 /
/

--
-- The same trend in the CO2 isotherm.
--
COPY
 'MLANG' 'MLANGSLV'  /
/

MULTIPLY
 'MLANGSLV'  1.4 /
/

LANGSOLV
 0.0        0.080
 279.       0.092
 661.       0.227
1000.       0.260
/

COALADS
 0.0  0.0   1.0
 0.5  0.5   0.2
 1.0  1.0   0.0
/

--
-- Connate gas saturation of 10 per cent
--
SWFN
  0.0    0.0   0.0
  0.2    0.0   0.0
  0.9    1.0   0.0
/

SGFN
  0.0     0.0   0.0
  0.1     0.0   0.0
  1.0     1.0   0.0
/

SSFN
 0     0  0
 0.01  0  0
 1     1  1
/
```

```
PVTW
  .0000000   1.00000    3.03E-06    .50000    0.00E-01 /

--   PGAS   BGAS   VISGAS
PVDG
      400    5.9   0.013
      800    2.95  0.0135
     1200    1.96  0.014
     1600    1.47  0.0145
     2000    1.18  0.015
     2400    0.98  0.0155
     2800    0.84  0.016
     3200    0.74  0.0165
     3600    0.65  0.017
     4000    0.59  0.0175
     4400    0.54  0.018
     4800    0.49  0.0185
     5200    0.45  0.019
     5600    0.42  0.0195 /

--   PGAS   BGAS   VISGAS
PVDS
      400    5.9   0.013
      800    2.95  0.0135
     1200    1.96  0.014
     1600    1.47  0.0145
     2000    1.18  0.015
     2400    0.98  0.0155
     2800    0.84  0.016
     3200    0.74  0.0165
     3600    0.65  0.017
     4000    0.59  0.0175
     4400    0.54  0.018
     4800    0.49  0.0185
     5200    0.45  0.019
     5600    0.42  0.0195 /

ROCK
 4000.00        .30E-05 /

DENSITY
 52.0000  64.0000   .04400 /

SDENSITY
 0.3   /

RPTPROPS
 'SOF2' 'SWFN' 'SGFN' 'PVTO'
 'PVTW' 'PVTG' 'DENSITY'  'ROCK'  'DIFFC'
 'LANGMUIR'  /

REGIONS

EQUALS
 'FIPNUM' 1  1 8 1 8 1 1 /
 'FIPNUM' 2  1 8 1 8 2 2/
/
```

SOLUTION

Eclipse 100 CO_2 injection example in the `SOLUTION` section:

```
SOLUTION  ==========================================================
EQUIL
 4015 528 3000 0 /
--
-- Initial 40 per cent of the fracture gas is CO2
--
SOLVFRAC
 128*0.4 /
RPTSOL
```

```
 'PRES'  'SWAT'  'SGAS'  'FIP=2' 'EQUIL'  'SSOL'
 'FIPSOL' 'GASCONC'  /
```

SUMMARY

Eclipse 100 CO_2 injection example in the `SUMMARY` section:

```
SUMMARY    ==========================================================
FPR
FWGR
FGPR
FWPR
FNPR
FNIR
FCGC
FCSC
RGIP
/
RCGC
 /
RCSC
 /
RNIP
 /
BCGC
 1 1 1 /
/
BCSC
 1 1 1 /
/
WBHP
/
WWGR
/
WGPR
/
WNPR
/
RUNSUM
SEPARATE
```

SCHEDULE

Eclipse 100 CO_2 injection example in the `SCHEDULE` section:

```
SCHEDULE  ==========================================================
RPTSCHED
 'PRES'  'SWAT'  'SGAS'  'RESTART=2' 'FIP=2'
 'WELLS=2' 'CPU=2' 'NEWTON=2' 'SSOL'
 'FIPSOL' 'GASCONC'  /
--
-- Initial period with no production
--
TSTEP
 1 9 90 /
WELSPECS
'P'  'G'   8  8  4000  'GAS'  /
'I'  'G'   1  1  4000  'GAS'  /
/
COMPDAT
'P'    2*   2   2 'OPEN'   0  .000000  .5000  .00000   .0000 0.000E-01 /
'I'    2*   2   2 'OPEN'   0  .000000  .5000  .00000   .0000 0.000E-01 /
/
WCONPROD
'P'  'OPEN'  'WRAT'     1*         1000.0      100000.00000
     1*          1*      50.000  0.000000   1*  0.00000000/
/
WCONINJE
 'I'  'GAS'  'SHUT'  'RATE'  1000.0  /
/
```

```
--
-- Initial production
--
NEXTSTEP
 0.1 /
TSTEP
 1.0 9.0 90.0 900.0 1000.0
/
--
-- Inject Co2
--
WCONINJE
 'I'  'GAS'  'OPEN'  'RATE'  100000.0  /
/

WSOLVENT
 'I' 1.0 /
/
NEXTSTEP
 0.1 /
TSTEP
 1.0 9.0 90.0 900.00 3*1000
/
--
-- Blow down
--
WCONPROD
'P'  'OPEN'  'GRAT'     1*         1000.0      100000.00000
     1*           1*      50.000  0.000000   1*  0.00000000/
/
WCONINJE
 'I'  'GAS'  'SHUT'  'RATE'  100000.0  /
/
NEXTSTEP
 0.1 /
TSTEP
 1.0 9.0 90.0 900.00 3*1000
/
END
```

Summary of keywords (Eclipse 300)

RUNSPEC section

Keyword	Description
COAL	Specifies that the coal bed methane model is to be used.
CBMOPTS	Options for coal bed methane
DUALPORO	Specifies a dual porosity run
NMATRIX	Define the number of matrix cells (multi-porosity option)

GRID section

Keyword	Description
DIFFMMF	Modify the matrix-fracture or matrix-matrix diffusivity.
SIGMA	Input the dual/multi-porosity SIGMA factor.
SIGMAV	Input the SIGMA factor on a cell by cell basis.
COALNUM	Assign coal regions
COALNUMR	Assign coal regions for matrix sub cells

Keyword	Description
CRNDENS	Input rock (coal) density by coal region number
ROCKDEN	Input rock (coal) density.
ROCKFRAC	Input rock (coal) fractions of the bulk volume on a cell by cell basis.
RPTGRID	Control output from the GRID section: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.
RPTGRIDL	Control output from the GRID section for local refined grids: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.
RPTINIT	Control output from the GRID and EDIT sections to INIT file: COALV and PORVF output pore volumes of the coal and fluid parts of grid cells.

PROPS section

Keyword	Description
LANGMEXT	Input the Langmuir volume and pressure constant for each component.
LANGMPL	Inputs a pressure scaling factor for the adsorption capacity for each grid cell.
LANGMUIR	Input the Langmuir table for each component.
DIFFCBM	Input gas diffusion data.
RESORB	Input gas readsorption factors.
LANGMULC	Input a scaling factor for the adsorption capacity for each grid cell and each component.
LANGMULT	Input a scaling factor for the adsorption capacity for each grid cell.
ROCKPAMA	Palmer-Mansoori rock compaction model.
ROCKPAME	Modified Palmer-Mansoori model made dependent on composition for coal swelling and shrinkage.
RPTPROPS	Controls output from the PROPS section. DIFFCBM outputs diffusion data. LANGMEXT outputs the extended Langmuir isotherm data. LANGMUIR outputs the Langmuir table data.
TPAMEPS	Component sorption induced strain values for the Palmer-Mansoori model
ROCKPAMA	This is an alternative to ROCKPAME.

SOLUTION section

Keyword	Description
GASADCO	Specify the initial concentration of adsorbed coal gas.
GASADEC	Specify the initial equilibrium concentration of adsorbed coal gas
GASCCMP	Specify the component used by GASCONC and GASSATC.
GASCONC	Specify the initial concentration of adsorbed coal gas for one component
GASSATC	Specify the initial equilibrium concentration of adsorbed coal gas for one component.
SORBFRAC	Scale initial adsorbed gas.

Keyword	Description
SORBPRES	Specify Langmuir sorption pressure.
RPTSOL	Controls output from the SOLUTION section. CGAS outputs the adsorbed gas concentration as surface volume / rock volume. The output array names are CGAS1, ..., CGASn (up to the number of components).

SUMMARY section

Refer to Miscellaneous well and group quantities in the *Eclipse Reference Manual.*

If methane is component number one, the methane production rate can be output by specifying:

```
FCWGPR
1 /
```

SCHEDULE section

Keyword	Description
WINJGAS	Specify nature of injection gas.
RPRKPAME	Report rock compaction multipliers (ROCKPAME) versus pressure for a given cell.
RPTSCHED / RPTRST	CGAS outputs the adsorbed gas concentration as surface volume / rock volume. The output array names are CGAS1, ..., CGASn (up to the number of components).

Using the Eclipse 300 coal bed methane model

The model is activated by specifying the keyword COAL in the RUNSPEC section. The dual porosity model should also be activated with the RUNSPEC keyword DUALPORO, and the number of coal regions should be set by item 6 of REGDIMS (default =1). For multi porosity cases, the keyword NMATRIX specifies the number of matrix cells per cell (see "Multi porosity"). It is possible to choose between two types of adsorption model: instant and time dependent. This is done by using the CBMOPTS keyword. For the instant adsorption model it is also possible to specify that no water flow is allowed in and out of a coal cell by using the CBMOPTS keyword.

Note: The coal bed methane option is currently not compatible with the Thermal option or the Eclipse 300 black oil option.

Grid data

Using the time dependent adsorption model and dual porosity, the first NDIVIZ /2 layers of the model represent the coal matrix and the second NDIVIZ /2 layers represent the fracture system. The input follows what is described for Eclipse 100 above, except:

- It is strongly recommended to use the keyword ROCKFRAC to set the coal volumes instead of specifying porosities when using the time dependent sorption model.
- The density of the coal must be supplied by the ROCKDEN keyword unless the LANGMUIR keyword is used.

- It is possible to set coal regions with the COALNUM keyword; the default is that all matrix cells are assigned to coal region number one.
- Using the multi-porosity option, the NMATOPTS keyword can be used to generate the pore volumes and coal volumes for the matrix sub-cells. This also generates SIGMA factors automatically for the partitioned matrix sub-grid cells based on the initial input sigma factor. Refer to "Multi porosity" and "Shale gas" for the definition of the coal and pore volume when doing matrix discretization.

Properties data

Gas and water PVT, EOS and relative permeability data need to be supplied in the usual way. In addition, the following keywords must be supplied:

Restriction	Keyword	Description
Required	LANGMEXT	Defines the Langmuir isotherm for each component and each coal region.
Required	LANGMUIR	Alternative to LANGMEXT, this keyword defines the Langmuir isotherm for each component and each coal region by table.
Required	DIFFCBM	Inputs the diffusion coefficients (not used by the instant sorption model)

If the GASWAT option is used water and gas relative permeability curves are entered by WSF and GSF. It is also possible to use the three phase relative permeability curves SWFN, SGFN and SOF3. However, in the case where there is no oil present, you must still enter an oil relative permeability curve. Optionally the FACTLI or ISGAS keywords can also be used in order to make sure that a single-phase hydrocarbon is not labeled as an oil phase during the simulation. Note that components predicted to be in an oil phase do not adsorb on the coal surface.

Note: If an OIL phase is present then using the option to interpolate the relative permeabilities around the critical point should be considered. This can be enabled using the argument KRMIX in the third item of the keyword CBMOPTS. The default behavior is not to perform the interpolation which is equivalent to using the NOMIX keyword.

The adsorbed coal gas is always reported as gas, where the density of the adsorbed gas is set to a liquid-like density.

Keyword	Description
RESORB	Inputs the re-adsorption factors (default=1.0)
LANGMULC	Inputs a scaling factor for the adsorption capacity for each grid cell and each component.
LANGMULT	Inputs a scaling factor for the adsorption capacity for each grid cell.
ROCKPAMA	Palmer-Mansoori rock compaction model.
ROCKPAME	Modified Palmer-Mansoori model made dependent on composition for coal swelling and shrinkage.

Equilibration

Time-dependent sorption

The matrix coal gas is always initialized according to the input matrix properties. This means that the input mole fractions and pressure are used to set the gas contents according to the Langmuir adsorption capacity.

An equilibrium is obtained if the initial specified composition of the fractures equals the input matrix composition and the system has free gas. In addition the pressure of the fracture must be level or higher than the input matrix pressure. If the fractures are fully water filled, an equilibrium mole fraction and sorption pressure is calculated internally.

Please note that the resulting adsorbed mole fractions generally are not equal to the input mole fractions.

When equilibration is done by EQUIL the fractures are equilibrated in the usual way. The initial coal gas molar densities are calculated from the prevailing matrix pressure and composition as specified by COMPVD or ZMFVD.

A different sorption pressure can be specified by the keyword SORBPRES. Entering pressure values which are lower than the computed matrix pressure will give a lower gas content. It is also possible to model undersaturated coal by scaling the calculated coal gas by a factor (see the keyword SORBFRAC).

It is also possible to input the initial coal gas concentration by using the keywords GASADCO and or GASADEC. The first specifies the concentration of the gas for each grid cell and each component. The second will scale the input Langmuir adsorption capacity so that the input concentration is matched at the given pressure and composition. It should be noted that the components need to be present, otherwise it is not possible to match the input concentrations. It is however possible to first input a composition used for scaling and thereafter use GASADCO to set the final concentration. It is also possible to select one component to use to obtain one scaling value by using the keyword GASSATC. In this case, all of the components use the scaling value obtained for this specific component.

Note that GASSATC, GASADEC, LANGMULT or LANGMULC cannot be used together.

An enumerated start is possible in the usual way. The input needed is:

- Mole fraction values for the gas phase (YMF) and oil phase (XMF). If no oil phase is present then values of zero (0) should be entered for XMF.
- Pressure.
- Gas and water saturations.

The matrix gas mole fractions and the pressure are then used to initialize the matrix coal gas.

Several equilibrium regions can be used in order to specify different conditions for the matrix and the fractures.

Instant sorption

For the instant sorption model the adsorbed gas content is calculated from the free gas phase composition and pressure for each simulation cell. If water is present the pressure in the pore space and the composition given by COMPVD or ZMFVD is used to determine the amount of adsorbed gas. SORBPRES and SORBFRAC can currently not be used with the instant sorption model.

Output and restart

For the time-dependent model the adsorbed molar densities are output in the usual way as molar densities in the coal volumes.

Using the instant desorption model the adsorbed molar density AMSC is also output, where MLSC are the molar densities in the pore volumes (AMSC is only output when a flexible restart file has been requested)

If component-specific scaling is applied, this is output as arrays MLANG1, ..., MLANGn. (Do not include LANGMULT values. If used, this is output to the INIT file as LANGMULT.)

The adsorbed gas concentration, reported as surface volume over rock volume, can be output by RPTRST with CGAS. The output array names are CGAS1,..., CGASn (up to the number of components).

Summary keywords

The SUMMARY keywords are:

Keyword	Description
FCWGPR	Production rate for a specific component (given in surface volume by assuming an ideal gas)
FCWGPT	Production total for a specific component (given in surface volume by assuming an ideal gas)
WYMF	Well mole fraction for a specific component

Examples

The following data sets come with the Eclipse installation:

File	Description
ECBM2.DATA	Enhanced coal bed methane example
ECBM2I.DATA	Instant sorption and single porosity
ECBM4.DATA	ECBM with coal shrinkage and swelling effects
CBMTRPL.DATA	Coal bed methane example using three porosities
CBM_E300_ROCKPAMA_TPAM.DATA	Coal bed methane example with Palmer-Mansoori rock compaction using TPAMEPS.

Table 7.1: Coal bed methane example files

Shale gas

	Eclipse 100
x	Eclipse 300

Shale gas reservoirs often have a matrix porosity system where the transient behavior in the matrix becomes important. Some of the gas might be adsorbed on the surface of the shale and some exists as a free gas in the matrix pore structure. In order to model such reservoirs the dual/multi porosity option can be used together with the Coal Bed Methane Model for adsorbed gas on the rock formation. See "Multi porosity" and "Adsorption models in Eclipse 300"

There are two models for flow of adsorbed gas into a simulation cell.

- Instant Sorption Model.

  The amount of adsorbed gas is in instant equilibrium with the free gas composition and pressure.

- Time-dependent Sorption Model.

  To choose between the Time-dependent or Instant Sorption Models the keyword `CBMOPTS` is used.

Simulating shale gas reservoirs

Grid cells with adsorption

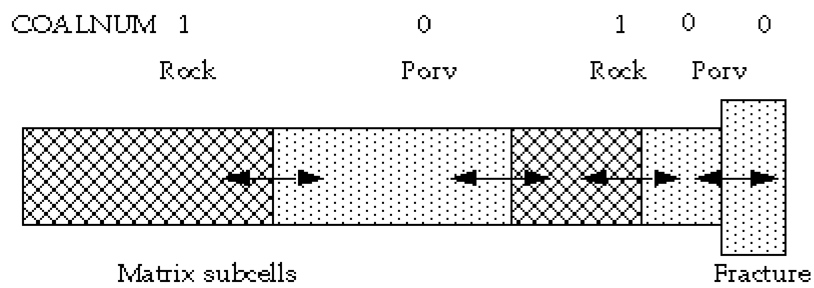
In order to specify cells having an adsorbed gas the `COALNUM` keyword is used for both methods. It is possible to redefine the coal region numbers by the keyword `COALNUMR`.

Time-dependent sorption

For the time-dependent method a simulation cell either contains free gas in a pore space or adsorbed gas in the rock. The rock is represented by one simulation cell and the pore volume by a connecting simulation cell. Darcy flow through a rock cell is not permitted. The usual pore volume of a cell now represents the rock volume where the imaginary micro pore space flow is accounted for by a diffusive flow equation. A cell having a non-zero coal region number as set by `COALNUM` hence needs to specify a “porosity” value that corresponds to a rock fraction value or using the keyword `ROCKFRAC`. For cells having a zero coal region number the porosity value correspond to the pore volume fraction.

Note: It is recommended that the `ROCKFRAC` keyword is used to set the rock volume and not the porosity keyword `PORO`.

If pore volume-pore volume connections exists a permeability value also need to be input in order to compute the transmissibility between the matrix subgrid cells. This is usually done by `PERMX`.



Matrix discretization (NMATOPTS) and time-dependent sorption

Using the time-dependent method the primary matrix subgrid cells are used to define both the total pore volume and the total coal volume that is partitioned. The primary matrix subgrid cells need to be defined as coal by COALNUM in order to tell that the matrix contains adsorbed gas. To redefine the matrix subgrid cells with a coal region number the keyword COALNUMR should be used.

With the time-dependent sorption model, the diffusive flow of adsorbed gas can be studied in detail using the matrix discretization option. The coal region numbers need to be the same for all connected matrix sub-grid cells. Mixing pore volume and coal volume with the option is not generally recommended as the interpretation of the resulting response from the matrix, which might include Darcy flow, needs to be carefully considered. Usually in such cases, the basic multi-porosity model should be used where the connection transmissibilities/diffusivities are controlled by the user-input sigma factors. Alternatively, the instant adsorption model can be used if appropriate. Darcy flow then controls the gas flowing towards the fractures within the pore space of the matrix.

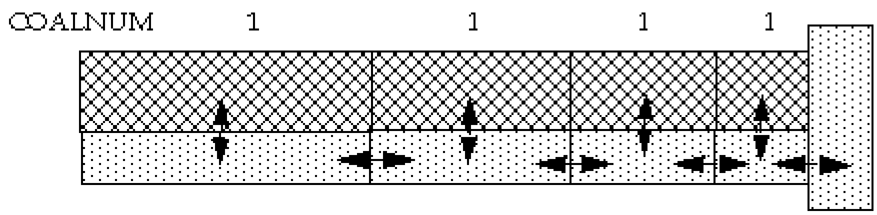
The matrix discretization option generates sigma factors for the connections. Thus for pore-volume to pore-volume connections, the permeabilities together with the generated sigma factor are used to generate the transmissibilities in a similar way as for dual porosity. It is possible to specify permeabilities individually for the different pore systems. If unspecified, the input permeability given to the primary matrix cell is used. The transmissibility would then be constructed according to the block volume of the sub-matrix grid cell and the permeability of the sub-matrix grid cell. Note that such a transmissibility must be viewed as an internal connection within the matrix block. The transport will be diffusive through a cell connected with a non-zero coal region number.

For more information about the diffusive flow see "Coal bed methane model".

The second item of NMATOPTS specifies a fraction of either the fracture pore volume or a fraction of the matrix block volume to be used for the first primary matrix grid cell. The method is selected by the third item of NMATOPTS. With the time-dependent method, it is usually better to specify a fraction of the matrix block volume since the first matrix sub-grid cell is not necessary a pore volume cell. Similarly if all the cells in the matrix are defined as rock cells (cells with a coal region number) there is no pore volume represented in the matrix. This means that if a non-zero pore volume is assigned to the primary matrix block on input, the volume will be neglected. Warnings are issued if this is the case. It should also be noted that if the third item of NMATOPTS is put to FPORV and there is no pore volume within the matrix, the partitioning method switches to use the MBLKV method for this matrix grid cell. Warnings are issued if this is the case. Any rock volume that is assigned to cells that are not coal cells will not alter the simulation results. Using the FPORV method with NMATOPTS, the first matrix block will get a size that will match the specified pore volume under the assumption that all the matrix blocks have a pore volume. However, if only some of the sub-matrix cells have a pore volume, a normalization of the input porosity is applied to these cells in order to match the input matrix grid cells total pore volume. As an example, if only one sub-matrix grid cell is a pore volume cell, the entire matrix pore volume is assigned to this sub-matrix grid cell.

Instant sorption

For the instant desorption model the pore volume has the usual interpretation, and the adsorption desorption process is represented as a source sink term into the matrix pore system. It should be noted that the coal volume of each matrix subgrid cell is computed from the average grid cell volume of each coal cell, where the pore volume of all the connected matrix and fracture grid cells are subtracted. If ROCKFRAC is specified for a non-coal cell, this volume is also subtracted from the coal volume. The coal volume is then distributed evenly between the coal cells.



Matrix discretization (NMATOPTS) and instant sorption

In combination with the multi porosity option the matrix subgrid cells are assigned pore volumes and rock/ coal volumes automatically if the NMATOPTS keyword is present. The primary (outer) matrix pore and rock volume is partitioned among the matrix subgrid cells. Note that the pore volume of the fractures are subtracted from the rock volume. The primary matrix subgrid cells need to be defined as coal by COALNUM in order to tell that the matrix contains adsorbed gas. To redefine the matrix subgrid cells with a coal region number the keyword COALNUMR should be used. Only the pore volume and coal volumes as defined by the primary matrix sub-grid cells are used to generate the matrix sub-grid cells coal and pore volumes.

Note that having different adsorption properties for the different subcells also means that the partitioning of the volumes between the subcells becomes important.

It is possible to assign a rock density for each coal region number by the keyword CRNDENS in order to assign different density to the matrix subgrid cells.

Coal and pore volumes

If the ROCKFRAC keyword is used we have the following pore volume and coal volume definitions for the primary matrix sub-grid cells:

Sorption model	Pore volume primary matrix subgrid, PV_m	Coal volume primary matrix subgrid, CV_m
Instant	$V_m \phi_m \times NTG_m$	$V_m r_m$
Time-dependent	$V_m \phi_m \times NTG_m$	$V_m r_m$

The following table shows the pore volume and coal volume definitions for the primary matrix subgrid cells if ROCKFRAC is not used.

Sorption model	Pore volume primary matrix subgrid, PV_m	Coal volume primary matrix subgrid, CV_m
Instant	$V_m \phi_m \times NTG_m$	$V_m - PV_m - PV_f$

Sorption model	Pore volume primary matrix subgrid, PV_m	Coal volume primary matrix subgrid, CV_m
Time-dependent	$V_m - CV_m - PV_f$	$V_m \phi_m \times NTG_m$

where

V_m, V_f Bulk volume, matrix or fracture

ϕ_m, ϕ_f Porosity, matrix or fracture

r_m Rock fraction value in matrix

NTG_m, NTG_f Net to gross, matrix or fracture

Examples

The following data sets come with the Eclipse installation:

File	Description
`MPORO.DATA`	Gas water case (matrix discretization)
`SHALEGAS1.DATA`	Shale gas
`SHALEGAS1T.DATA`	Shale gas with time-dependent sorption

Table 7.2: Shale gas example files